

Revisiting the calculation of PAL from physical activity data

Faidon Magkos 

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ABSTRACT

The physical activity level (PAL), defined as the ratio of total daily energy expenditure to resting metabolic rate, is widely used to assess energy requirements in both clinical and research settings. PAL is often not estimated directly—via measurements of total energy expenditure and resting metabolic rate—but rather indirectly using self-reported or objectively recorded physical activity data interpreted through the metabolic equivalent of task (MET) system. However, exercise physiology research consistently shows that the conventional value of 1 MET (set at 1 kcal/kg-hour, corresponding to a resting oxygen consumption of 3.5 mL/kg-min) overestimates true resting metabolic rate by 10%–30%. This systematic bias can lead to inaccurate PAL and energy expenditure estimates when relying solely on physical activity data. To address this problem, a simple correction formula is proposed for adjusting PAL values when individual resting metabolic rate is available or can be predicted from demographic and anthropometric data.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ The physical activity level (PAL) is commonly estimated from physical activity data using standard metabolic equivalent of task (MET) values; however, the conventional definition of 1 MET systematically overestimates true resting metabolic rate, potentially biasing PAL estimates.

WHAT THIS STUDY ADDS

⇒ This study shows that MET-derived PAL is systematically underestimated and introduces a simple correction using individual resting metabolic rate that aligns indirect PAL estimates with true PAL.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Applying this correction can improve the accuracy of energy requirement estimates in research and practice when resting metabolic rate can be measured or predicted.

Adequate physical activity is important for maintaining good health and avoiding disease. Reliably assessing physical activity, however, presents many challenges. Self-report (subjective) measures have traditionally been the method of choice for physical activity surveillance in most population-based studies; however, objective tools are increasingly being used.¹ Although subjective measures are not particularly informative at the individual level, when applied to adequately large groups (populations), the takeaway from self-reported physical activity is largely the same as that with objectively measured physical activity.²

Very often, subjective physical activity data are used together with the metabolic equivalent of task (MET) to make inferences about energy expenditure and study the link between physical activity and chronic disease.^{3–8} The MET is essentially a unit of relative energy expenditure, defined as the gross energy expenditure (measured by indirect calorimetry) during a particular activity that is typically performed in the non-fasted state,^{9–11} divided by the resting metabolic rate (RMR)—as such, it indicates the real-life energy cost of physical activity as a multiple

of RMR. By convention, 1 MET is defined as the rate of energy expenditure while resting quietly in a chair and equals 1 kcal/kg-hour ('reference RMR'), corresponding to a resting oxygen uptake of 3.5 mL/kg-min.¹² The MET factor system was initially devised for standardising different activities and classifying them into categories of intensity.¹² Subsequently, it evolved into a valuable tool for estimating physical activity energy expenditure, and not least for calculating the physical activity level (PAL).

The concept of PAL emerged in 1985 when the Food and Agriculture Organization, the WHO and the United Nations University jointly recommended that estimation of energy requirements be based on the assessment of energy expenditure rather than energy intake.¹³ PAL is defined as the ratio of total energy expenditure (TEE, which is the sum of RMR, thermic effect of food and physical activity energy expenditure) to RMR and indicates the overall level of physical activity of an individual, that is, how active or inactive their lifestyle is. As such, like MET, it is also expressed as a multiple of RMR but reflects the whole day. PAL provides a unitless, directly comparable metric of physical



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Department of Nutrition, Exercise and Sports, University of Copenhagen, Frederiksberg, Copenhagen, Denmark

Correspondence to

Dr Faidon Magkos;
fma@nexs.ku.dk

activity and allows for calculating the food energy an individual needs to consume to maintain their body weight and their specific lifestyle.¹⁴ This method of calculating energy requirements is referred to as the ‘factorial approach’, exactly because it is based on the MET factor system.¹³

PAL can be calculated either directly, by measuring TEE and RMR by available gold-standard techniques (eg, doubly labelled water or indirect calorimetry), or indirectly, based solely on physical activity data. The latter approach is more cost-effective, technically less demanding and scalable. It is preferred not only in epidemiological research but also by many exercise physiologists, nutritionists and medical professionals for estimating energy requirements.¹⁵

To calculate PAL from physical activity data, one only needs to know what activities an individual engages in during the day and for how long (columns a and b in [figure 1-top](#)). Then, one can assign the corresponding MET value to each activity, multiply by the duration, and sum for the whole day to derive the accumulated physical activity in standardised units of MET-by-time (MET-hours or MET-minutes; columns c and d in [figure 1-top](#)). PAL is computed by dividing the daily MET-hours by 24 (hours per day) or the daily MET-minutes by 1440 (minutes per day).^{3 4} This widely used method is solely based on physical activity data and negates the need for collecting demographic, anthropometric or metabolic data (ie, measuring or predicting RMR).

However, this approach to calculating PAL is not necessarily equivalent to calculating PAL as per its definition by using the direct method (ie, ‘true PAL’ = TEE/RMR). Using the reference RMR of 1 kcal/kg·hour, the MET-hours for each activity can be converted back to gross energy expenditure, summed up to calculate daily TEE (column e in [figure 1-top](#)), and then divided by the RMR of the individual to compute true PAL which, in the working example of [figure 1-top](#), is much higher. True PAL in this context refers to PAL calculated using the measured or predicted RMR for the individual, as opposed to using the reference RMR.

The validity of the standard MET approach will obviously depend on how accurate the physical activity data are, but also on how well the reference RMR of 1 MET=1 kcal/kg·hour reflects the RMR of any given individual (ie, the ‘true RMR’). The 1 MET convention systematically overestimates resting metabolism by 10%–15% and up to 30%, and this is not merely an issue of sitting (reference RMR) versus lying (true RMR), or of measuring versus predicting the true RMR.^{16–18} Likewise, large cohort studies (n=430–769) report average resting oxygen uptake values of 2.6–3.0 mL/kg·min, that is, 15%–25% lower than the 3.5 mL/kg·min standard.^{16 19} The bias is greater among those with higher body mass index and lower fitness.^{16 17 20} It is noteworthy that the origin of the MET convention remains obscure, likely stemming from early investigations involving one or a very limited number of male subjects.¹⁶

The result of these discrepancies is that standard MET factors underestimate true, ‘personalised’ MET values calculated by using the true RMR (pMET, column f in [figure 1-top](#)).^{16 20} If one uses these corrected pMET values to recalculate daily pMET-hours, the PAL derived by the indirect approach is the same as the true PAL calculated by the direct approach, that is, as TEE/RMR (bottom of columns e and g in [figure 1-top](#)).

An appropriate correction must, therefore, be applied to PAL values estimated by the standard MET approach, taking into account the degree of agreement between the reference and the true RMR ([figure 1-middle](#)). Although numerous factors—such as age, sex, body mass index, body composition and genetic variability—can influence RMR, the applied correction is designed to improve the accuracy of the PAL estimate at the individual level. This simple correction involves dividing the reference RMR by the true RMR (measured or predicted). Accordingly, factors affecting RMR are not expected to have any bearing on the outcome of the correction for a given individual. Further, it can be demonstrated that the true PAL is a function of both the magnitude of error between the reference and the true RMR, and the level of physical activity as assessed by the standard MET approach—this can be visualised by the progressively steeper slopes with increasing standard PAL values ([figure 1-bottom](#)).

The mean PAL value obtained by the standard MET approach in the general population in the USA and the UK, using self-reported physical activity data, is approximately 1.60 (ranging from 1.59 to 1.63), with a SD of about 0.25 (equivalent to a 95% range of 1.1–2.1).^{3 4} The mean PAL value calculated as the ratio between TEE measured by doubly labelled water and RMR measured by indirect calorimetry in representative samples from multiple countries is approximately 1.74 (ranging from 1.65 to 1.85), with a SD of about 0.20 (equivalent to a 95% range of 1.35–2.15).^{21 22} These observations indicate that population PAL estimates from the indirect method (based on physical activity data and the MET system) are lower than population PAL estimates from the direct method (based on measured TEE/RMR), which could well be due to the bias of the MET convention.

Moreover, self-report tools often overestimate physical activity and associated energy expenditure compared with accelerometry.² This overestimation counterbalances some of the underestimation caused by using the standard MET approach and is likely the reason why self-report tools provide reasonable approximations of PAL at the population level.^{3 4} However, in a scientific field that strives for measurement accuracy, it is hardly acceptable to keep relying on methods that yield merely reasonable results at the group level because of compounding errors that are in opposite directions. In circumstances where self-reported physical activity is underestimated,²³ the error in estimating PAL and TEE will be even greater (shorter duration multiplied by lower MET).

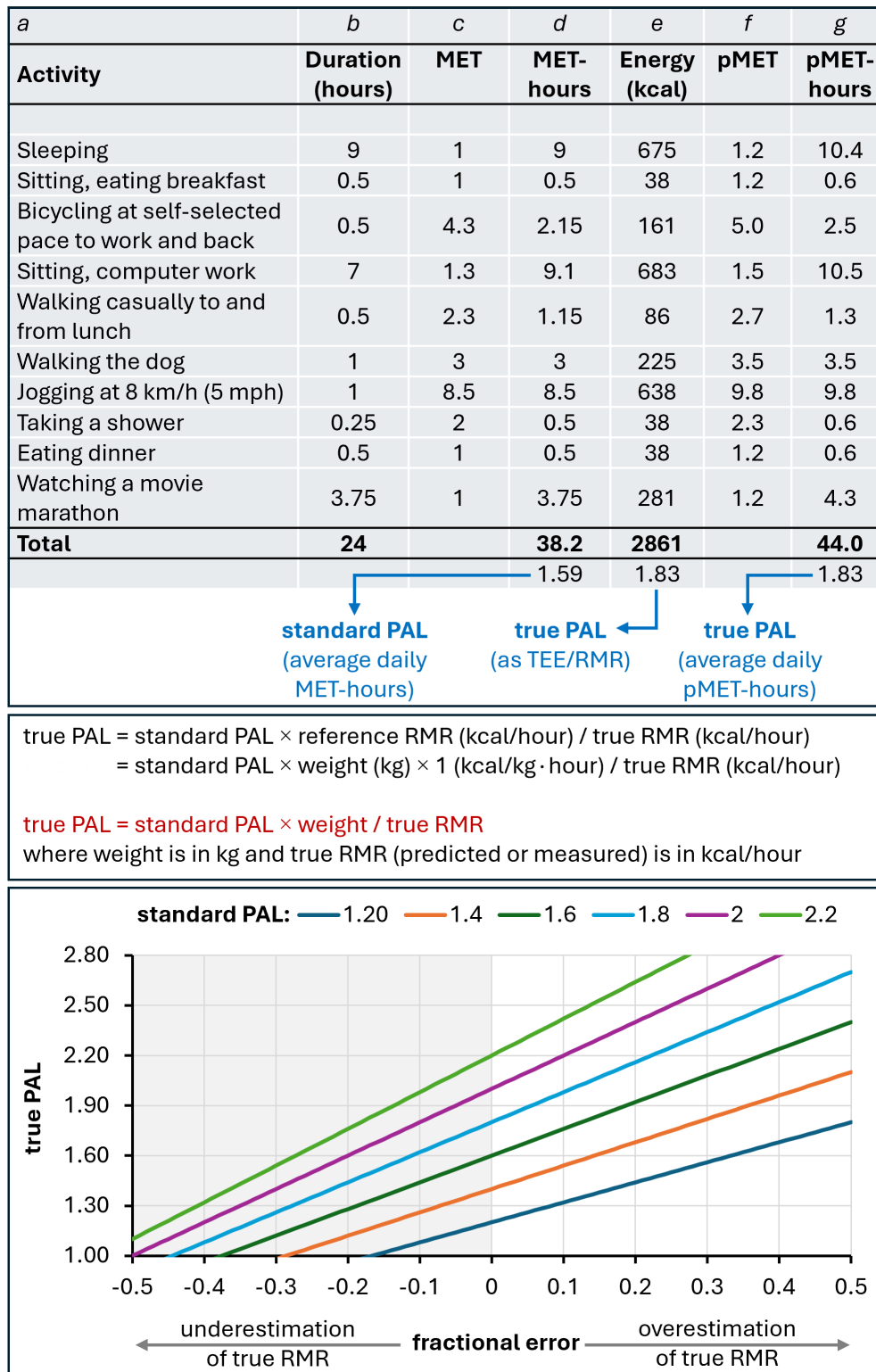


Figure 1 Top: example of using the standard MET and the personalised MET (pMET) approach to calculate PAL for an individual weighing 75 kg with a true RMR of 1560 kcal/day (65 kcal/h), who reported their physical activities over the course of one day. The standard MET values for each activity were derived from the Compendium of Physical Activities.¹² pMET values were calculated according to Byrne *et al.*¹⁶ Middle: derivation of a simple formula to correct PAL values estimated by the standard MET approach. Bottom: dependence of true PAL on the difference between the reference RMR and the true RMR. The fractional error was calculated as: (reference RMR – true RMR) / true RMR. Positive errors indicate that the reference RMR of 1 MET overestimates the true RMR (very likely), whereas negative values indicate an underestimation of the true RMR (not likely; shaded area). The linear function for these lines is: true PAL = standard PAL × (fe + 1), where fe is the fractional error. kcal, kilocalories; MET, metabolic equivalent of task; PAL, physical activity level; pMET, personalised MET; RMR, resting metabolic rate; TEE, total energy expenditure.

The bias in question is not confined to self-report instruments; rather, it extends to all methods of physical activity monitoring, including objective tools, to the extent these tools misrepresent physical activity energy expenditure (not necessarily its duration). Accelerometers, for example, quantify locomotion, which is then converted into time spent on physical activities of varying intensity (sedentary, low, moderate and vigorous). Classification of intensity is done by assigning specific thresholds for counts-per-minute (the basic unit of accelerometry data), which are validated against standard MET factors.²⁴ Subsequent conversion into familiar units of energy expenditure requires additional calibration—a process on which there is little consensus and much variability across devices.²⁴ Not surprisingly, therefore, measurement errors in the energy cost of daily physical activity above resting levels for common research-grade accelerometers are on the order of 24% at the group level, and 50%–60% at the individual level.²⁵ Accordingly, the calculation of PAL from objectively measured physical activity will not necessarily be more accurate than that from self-report tools.

As a final remark, the arguments presented in this perspective are not intended to diminish the value of the MET system but rather to refine one of its applications for purposes beyond its original design. Indeed, all versions of the Compendium of Physical Activities explicitly state that MET values are not meant for precise quantification of energy expenditure or PAL but serve to standardise and compare the relative intensity of various physical activities.^{12 17}

In conclusion, in studies where the PAL is used to convert physical activity data to energy expenditure for evaluating links with disease or estimating energy requirements, it is recommended that the correction provided herein be applied to PAL values calculated by the standard MET approach, at least whenever basic demographic and anthropometric data are available to predict RMR. Otherwise, results should be interpreted with caution.

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ORCID iD

Faidon Magkos <https://orcid.org/0000-0002-1312-7364>

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